Lecture 4

The free particle: observables without eigenvalues

Objectives. *The Fourier transform builds the position, momentum and free energy on the line. These observables are self-adjoint and have no eigenvalues. For every unit vector in $\mathcal{S}(\mathbb{R})$, we prove uncertainty with equality exactly for the Gaussian packets. In a box the same inequality fails for a self-adjoint momentum P_θ, at its eigenvectors.*

- Compute a Gaussian packet's momentum distribution.
- State the domains needed for the canonical commutation relation.
- Derive the boundary term in the interval uncertainty inequality.

4.1 From the circle to the line

Take the circle of § 3.1, with circumference L, and place it on the interval $[-L/2, L/2]$. Its momentum has the eigenvalues $p_k = 2\pi\hbar k/L$, for $k \in \mathbb{Z}$, with the eigenvectors $f_k(x) = e^{ip_k x/\hbar}/\sqrt{L}$. Neighbouring eigenvalues are $\Delta p = 2\pi\hbar/L$ apart. Let ψ be a smooth function that vanishes outside an interval $[-R, R]$, and let $L > 2R$. Since ψ vanishes outside $[-R, R]$, the integral over the circle is an integral over the line, and the coefficients of ψ are

$$c_k = \langle f_k, \psi \rangle = \frac{1}{\sqrt{L}} \int_{-\infty}^{\infty} e^{-ip_k x/\hbar} \psi(x)\, dx = \left(\frac{2\pi\hbar}{L}\right)^{1/2} \tilde{\psi}(p_k),$$

where the *Fourier transform* $\tilde{\psi}$ of ψ is, for every real p,

$$\tilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} e^{-ipx/\hbar} \psi(x)\, dx. \tag{4.1}$$

By the standing fact S2, shifted to $[-L/2, L/2]$, the vectors f_k form an orthonormal basis. So Parseval's identity and the formula for c_k give

$$\|\psi\|^2 = \sum_{k\in\mathbb{Z}} |c_k|^2 = \sum_{k\in\mathbb{Z}} \Delta p \, \left|\tilde{\psi}(p_k)\right|^2 .$$

As L grows, the spacing Δp tends to 0, and the sum looks like a Riemann sum of $\int \left|\tilde{\psi}(p)\right|^2 dp$. The eigenvalues of the momentum fill the line, and the numbers $c_k/\sqrt{\Delta p} = \tilde{\psi}(p_k)$ sample one function $\tilde{\psi}$ ever more finely. The Riemann sum suggests that the transform keeps norms on the line. It does not prove it, and Theorem 4.2 below states the fact.

Figure 4.1 draws these samples for one state at $\hbar = 1$. The state is $\psi(x) = Ce^{-1/(1-x^2)}$ for $|x| < 1$ and $\psi(x) = 0$ otherwise, with $C > 0$ chosen so that $\|\psi\| = 1$. It is smooth and vanishes outside $[-1, 1]$, so $R = 1$, and the circles have $L = 4$, 8 and 16. Each doubling

of L halves the spacing Δp. The values $|c_k|^2/\Delta p = \left|\tilde{\psi}(p_k)\right|^2$ lie on the one curve $\left|\tilde{\psi}\right|^2$, closer together as L grows.

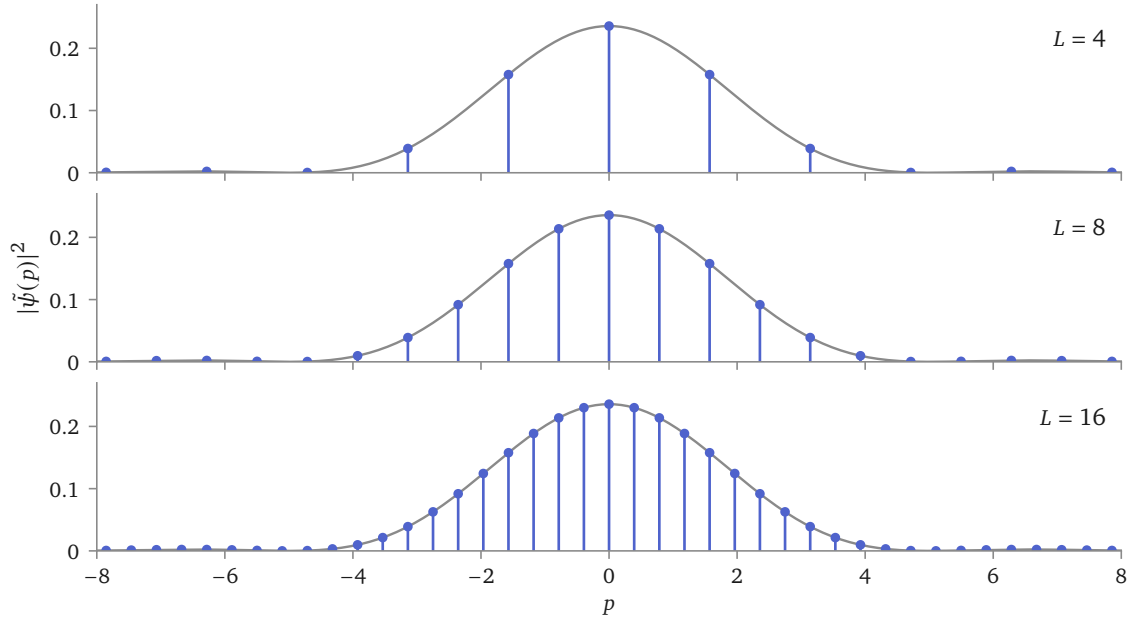


Figure 4.1: The squared transform $\left|\tilde{\psi}(p)\right|^2$ (grey, the same in every row) and, for each circumference L, a stem of height $|c_k|^2/\Delta p$ at each eigenvalue $p_k = 2\pi k/L$, at $\hbar = 1$. A stem is the probability $|c_k|^2$ of the reading p_k of the circle's momentum, divided by the spacing Δp, and it ends on the curve.

The formula (4.1) needs ψ to decrease fast enough for the integral to converge. The following functions do, and so do their derivatives.

Definition 4.1 (Rapidly decreasing functions)**.** The *Schwartz space* $\mathcal{S}(\mathbb{R})$ is the set of infinitely differentiable functions $\psi : \mathbb{R} \to \mathbb{C}$ such that

$$x^a \psi^{(b)}(x) \text{ is bounded on } \mathbb{R} \quad \text{for all integers } a, b \geq 0.$$

For example, every Gaussian $e^{-(x-q)^2/4s^2 + ip_0 x/\hbar}$, with $s > 0$, lies in $\mathcal{S}(\mathbb{R})$, and so does its product with any polynomial, since each derivative is again a polynomial times the Gaussian. So does every smooth function that vanishes outside a bounded interval. The function $1/(1+x^2)$ is square-integrable, but it is not in $\mathcal{S}(\mathbb{R})$, since $x^3/(1+x^2)$ is unbounded.

By the product rule, $\mathcal{S}(\mathbb{R})$ is a subspace that multiplication by x and differentiation map into itself. Let $\psi \in \mathcal{S}(\mathbb{R})$. Since ψ and $x^2\psi$ are bounded, there is a constant C with $(1+x^2)\,|\psi(x)| \leq C$, so $|\psi(x)| \leq C/(1+x^2)$. Hence ψ is integrable, since $\int dx/(1+x^2) = \pi$, and square-integrable, since $|\psi(x)|^2 \leq C^2/(1+x^2)$. So the integral (4.1) converges for every p. By the standing fact S4, $\mathcal{S}(\mathbb{R})$ is dense in $L^2(\mathbb{R})$.

Theorem 4.2 (Plancherel · *cited*)**.** *For every $\psi \in \mathcal{S}(\mathbb{R})$, $\int \left|\tilde{\psi}(p)\right|^2 dp = \int |\psi(x)|^2\, dx$. The map $\psi \mapsto \tilde{\psi}$ extends to a unitary operator F on $L^2(\mathbb{R})$, and its inverse F^{-1} is the same construction with $e^{+ipx/\hbar}$ in place of $e^{-ipx/\hbar}$.*

This is the standing fact S1 (*Linear Algebra*, Lecture 13; Hall, Thms A.18 and A.19, whose

transform is ours with $\hbar = 1$). The transform of a derivative is computed by integration by parts.

Proposition 4.3 (The transform of a derivative). *For every $\psi \in \mathcal{S}(\mathbb{R})$, the transform of ψ' is $(\mathrm{i}p/\hbar)\tilde{\psi}(p)$. So the transforms of $-\mathrm{i}\hbar\psi'$ and of $-(\hbar^2/2m)\psi''$ are $p\tilde{\psi}(p)$ and $(p^2/2m)\tilde{\psi}(p)$.*

Proof. By integration by parts on $[-R, R]$,

$$\int_{-R}^{R} e^{-\mathrm{i}px/\hbar}\psi'(x)\,\mathrm{d}x = \left[e^{-\mathrm{i}px/\hbar}\psi(x)\right]_{-R}^{R} + \frac{\mathrm{i}p}{\hbar}\int_{-R}^{R} e^{-\mathrm{i}px/\hbar}\psi(x)\,\mathrm{d}x.$$

As $R \to \infty$, the bracket tends to 0, since $\psi(\pm R) \to 0$, and both integrals converge, since ψ and ψ' are integrable. Dividing by $\sqrt{2\pi\hbar}$ gives the transform of ψ'. Applying the rule to $\psi' \in \mathcal{S}(\mathbb{R})$ gives the transform $(\mathrm{i}p/\hbar)^2\tilde{\psi}$ of ψ''. The last two claims follow, since $-\mathrm{i}\hbar(\mathrm{i}p/\hbar) = p$ and $-(\hbar^2/2m)(\mathrm{i}p/\hbar)^2 = p^2/2m$. ∎

In words, the transform turns the rule $-\mathrm{i}\hbar\,\mathrm{d}/\mathrm{d}x$ into multiplication by p, as the circle's coefficients turned the momentum into multiplication by p_k.

Self-check 4.1

1. Let $\psi \in \mathcal{S}(\mathbb{R})$, and let a and b be real. Show that the transform of $\psi(x-a)$ is $e^{-\mathrm{i}pa/\hbar}\tilde{\psi}(p)$, and that the transform of $e^{\mathrm{i}bx/\hbar}\psi(x)$ is $\tilde{\psi}(p-b)$.

4.2 Position and momentum

Definition 4.4 (Position, momentum and free energy on the line). On $L^2(\mathbb{R})$, the *position* Q, the *momentum* P and the *free energy* H_0 of a particle of mass m have the domains and actions

$$\begin{aligned} D(Q) &= \{\psi :\ x\psi \in L^2(\mathbb{R})\}, & (Q\psi)(x) &= x\psi(x),\\ D(P) &= \{\psi :\ p\tilde{\psi} \in L^2(\mathbb{R})\}, & P\psi &= F^{-1}\big(p\tilde{\psi}\big),\\ D(H_0) &= \{\psi :\ p^2\tilde{\psi} \in L^2(\mathbb{R})\}, & H_0\psi &= F^{-1}\Big(\frac{p^2}{2m}\tilde{\psi}\Big). \end{aligned}$$

Here $\tilde{\psi}$ stands for $F\psi$, for every $\psi \in L^2(\mathbb{R})$. In words, P and H_0 multiply the transform by p and by $p^2/2m$.

Postulate 4.5 (A particle on a line). The states of a particle on a line are the rays of $L^2(\mathbb{R})$. Its position and momentum are the observables Q and P, and if it is free, its energy is the observable H_0. An ideal reading of "Q lies in I", for an interval I, follows the rule of Postulate 2.6 with the projection 1_I. An ideal reading of "P lies in I" follows it with the projection $F^{-1}1_I F$.

Let ψ be a unit vector. By the rule of Postulate 2.6 with the projection $F^{-1}1_I F$, and since F is unitary, a momentum reading lies in I with the probability

$$\mathrm{Prob}(P \in I) = \left\|F^{-1}1_I F\psi\right\|^2 = \left\|1_I\tilde{\psi}\right\|^2 = \int_I \left|\tilde{\psi}(p)\right|^2 \mathrm{d}p.$$

So $|\psi(x)|^2$ is the probability density of position readings, and $\left|\tilde{\psi}(p)\right|^2$ that of momentum readings. Lecture 5 derives both projections from one rule for every self-adjoint operator.

The operators of Definition 4.4 are built from multiplication operators. Recall the two facts about them that § 0.4 proves. By Lemma 0.7, multiplication M_g by a real continuous function g, on $D(M_g) = \{\psi : g\psi \in L^2(\mathbb{R})\}$, is self-adjoint, and a value that g takes at only finitely many points is not an eigenvalue. By Lemma 0.8, $U^{-1}AU$ is self-adjoint, with the eigenvalues of A, when U is unitary and A is self-adjoint.

Theorem 4.6 (Position, momentum and free energy)**.** *The operators Q, P and H_0 are self-adjoint, and none of them has an eigenvalue. Every $\psi \in \mathcal{S}(\mathbb{R})$ lies in their domains, and*

$$Q\psi = x\psi, \qquad P\psi = -\mathrm{i}\hbar\psi', \qquad H_0\psi = -\frac{\hbar^2}{2m}\psi''.$$

Moreover $H_0 = P^2/2m$, where $P^2 = PP$ acts on $\{\psi \in D(P) : P\psi \in D(P)\}$.

Proof. First, $Q = M_x$, and x takes each value once. By Lemma 0.7, Q is self-adjoint with no eigenvalue. Next, $P = F^{-1}M_pF$ and $H_0 = F^{-1}M_{p^2/2m}F$, and $p^2/2m$ takes each value at most twice. By Lemmas 0.7 and 0.8, both are self-adjoint with no eigenvalue. For $\psi \in \mathcal{S}(\mathbb{R})$, $x\psi$ lies in $\mathcal{S}(\mathbb{R})$, so in $L^2(\mathbb{R})$. By Proposition 4.3, $p\tilde{\psi}$ is the transform of $-\mathrm{i}\hbar\psi' \in \mathcal{S}(\mathbb{R})$, so it lies in $L^2(\mathbb{R})$, and applying F^{-1} gives $P\psi = -\mathrm{i}\hbar\psi'$. The same argument with $-(\hbar^2/2m)\psi''$ gives H_0. Finally, $F(P\psi) = p\tilde{\psi}$, so $\psi \in D(P^2)$ exactly when $p\tilde{\psi}$ and $p^2\tilde{\psi}$ are square-integrable. Since $|p| \le 1 + p^2$, the second implies the first. So $D(P^2) = D(H_0)$, and $P^2\psi = F^{-1}(p^2\tilde{\psi}) = 2mH_0\psi$. ∎

None of the three observables has an eigenvector, so none has an eigenbasis, and the rule for readings of Postulate 1.5 does not apply to them. By Postulate 4.5, their readings are made on intervals.

The plane wave $e^{\mathrm{i}p_0x/\hbar}$ satisfies $-\mathrm{i}\hbar\psi' = p_0\psi$ at every point, and physics texts call it a momentum eigenstate. But $\left|e^{\mathrm{i}p_0x/\hbar}\right| = 1$, so the integral of its square modulus is infinite. It is not in $L^2(\mathbb{R})$, and it is not a state. No state has a sharp momentum. Lecture 6 shows in which sense longer and longer packets approach the plane wave.

Self-check 4.2

1. Is $\psi(x) = (1+x^2)^{-3/4}$ in $L^2(\mathbb{R})$? Is it in $D(Q)$?

4.3 The free energy and the canonical relation

By Propositions 3.11 and 3.13, in a box no self-adjoint momentum squares to the energy. On the line the momentum does, since $H_0 = P^2/2m$ by Theorem 4.6. Let $\psi \in D(H_0)$. The proof of Theorem 4.6 showed that $D(H_0) = D(P^2)$, which lies in $D(P)$. So, since F is unitary,

$$\langle\psi, H_0\psi\rangle = \int_{-\infty}^{\infty} \frac{p^2}{2m}\left|\tilde{\psi}(p)\right|^2 \mathrm{d}p = \frac{\|P\psi\|^2}{2m} \ge 0.$$

Lecture 6 defines the *spectrum* of an operator, the set that takes the place of its eigenvalues, and shows that the spectrum of H_0 is $[0, \infty)$.

The canonical relation $QP - PQ = \mathrm{i}\hbar$ cannot hold for $n \times n$ matrices (*Quantum Mechanics I*, Prop. 9.8), nor for bounded self-adjoint operators (§ 0.1). *Quantum Mechanics I* promised it "on a suitable set of states". Here is the set.

Proposition 4.7 (The canonical commutation relation)**.** *The operators Q and P map $\mathcal{S}(\mathbb{R})$ into itself, and*

$$QP\psi - PQ\psi = \mathrm{i}\hbar\psi \tag{4.2}$$

for every $\psi \in \mathcal{S}(\mathbb{R})$. The relation (4.2) holds for every $\psi \in D(QP) \cap D(PQ)$, where $D(QP) = \{\psi \in D(P) : P\psi \in D(Q)\}$ and $D(PQ) = \{\psi \in D(Q) : Q\psi \in D(P)\}$.

Proof. On $\mathcal{S}(\mathbb{R})$, Q and P act by x and by $-\mathrm{i}\hbar\,\mathrm{d}/\mathrm{d}x$ (Theorem 4.6), which map $\mathcal{S}(\mathbb{R})$ into itself. By the product rule,

$$QP\psi - PQ\psi = -\mathrm{i}\hbar x\psi' + \mathrm{i}\hbar(\psi + x\psi') = \mathrm{i}\hbar\psi.$$

This computation is *Linear Algebra*, Thm 14.15. Next, let $\psi \in D(QP) \cap D(PQ)$ and $\varphi \in \mathcal{S}(\mathbb{R})$. Since Q and P are symmetric, and $Q\varphi$ and $P\varphi$ lie in $\mathcal{S}(\mathbb{R})$,

$$\langle \varphi, QP\psi\rangle = \langle Q\varphi, P\psi\rangle = \langle PQ\varphi, \psi\rangle\,, \qquad \langle \varphi, PQ\psi\rangle = \langle P\varphi, Q\psi\rangle = \langle QP\varphi, \psi\rangle\,.$$

Subtract the second identity from the first. By (4.2) for φ, and since the inner product is antilinear in its first slot,

$$\langle \varphi, QP\psi - PQ\psi\rangle = \langle PQ\varphi - QP\varphi, \psi\rangle = \langle -\mathrm{i}\hbar\varphi, \psi\rangle = \langle \varphi, \mathrm{i}\hbar\psi\rangle\,.$$

So the vector $v = QP\psi - PQ\psi - \mathrm{i}\hbar\psi$ is orthogonal to the dense set $\mathcal{S}(\mathbb{R})$. By the argument of § 0.2 for the uniqueness of the adjoint, $v = 0$. ∎

The relation needs both products, so it is a statement about $D(QP) \cap D(PQ)$, not about all of $D(Q) \cap D(P)$.

Self-check 4.3

1. For $\psi(x) = (2\pi s^2)^{-1/4} e^{-x^2/4s^2}$, compute $\langle \psi, H_0\psi\rangle = \|P\psi\|^2/2m$. How small can it be made? Can the mean energy of a state be 0?

4.4 The uncertainty relation

Let A be an operator and $\psi \in D(A)$ a unit vector with a real mean $\langle A\rangle = \langle \psi, A\psi\rangle$. The *spread* of A in ψ is

$$\Delta A = \|(A - \langle A\rangle)\psi\|\,.$$

For an observable with an eigenbasis, this is the spread of its readings, as in *Quantum Mechanics I*, Lecture 5. For Q, ΔQ is the spread of the density $|\psi|^2$, since

$$(\Delta Q)^2 = \int_{-\infty}^{\infty} (x - \langle Q\rangle)^2\, |\psi(x)|^2\, \mathrm{d}x.$$

For P, by the unitarity of F, ΔP is the spread of the density $|\tilde{\psi}|^2$. As in § 2.6, we write $\Delta A = \infty$ when $\psi \notin D(A)$.

Lemma 4.8 (Robertson's inequality for symmetric operators). *Let A and B be symmetric and ψ a unit vector in $D(A) \cap D(B)$. Then*

$$\Delta A\, \Delta B \geq \tfrac{1}{2}\,|\langle A\psi, B\psi\rangle - \langle B\psi, A\psi\rangle|\,.$$

Proof. Put $\tilde{A} = A - \langle A\rangle$ and $\tilde{B} = B - \langle B\rangle$. Since A and B are symmetric, their means are real, by a self-check of § 0.2. Since also $\|\psi\| = 1$, expanding gives

$$\big\langle \tilde{A}\psi, \tilde{B}\psi\big\rangle = \langle A\psi, B\psi\rangle - \langle A\rangle\langle B\rangle \quad \text{and} \quad \big\langle \tilde{B}\psi, \tilde{A}\psi\big\rangle = \langle B\psi, A\psi\rangle - \langle A\rangle\langle B\rangle.$$

Write $z = \big\langle \tilde{A}\psi, \tilde{B}\psi\big\rangle$, so that $\big\langle \tilde{B}\psi, \tilde{A}\psi\big\rangle = \overline{z}$. Subtracting the two identities gives $\langle A\psi, B\psi\rangle - \langle B\psi, A\psi\rangle = z - \overline{z} = 2\mathrm{i}\,\mathrm{Im}\,z$. Its modulus is $2\,|\mathrm{Im}\,z| \leq 2\,|z|$, and by Cauchy–Schwarz $|z| \leq \big\|\tilde{A}\psi\big\|\,\big\|\tilde{B}\psi\big\| = \Delta A\,\Delta B$. ∎

This is *Quantum Mechanics I*, Thm 5.7, for a pure state, with operators that need not act on every vector.

Theorem 4.9 (The uncertainty relation). *For every unit vector $\psi \in \mathcal{S}(\mathbb{R})$,*

$$\Delta Q\, \Delta P \geq \frac{\hbar}{2}.$$

Equality holds if and only if

$$\psi(x) = c\, \exp\Big(-\frac{(x-q)^2}{4s^2} + \frac{\mathrm{i}p_0 x}{\hbar}\Big) \tag{4.3}$$

for real numbers q and p_0, a width $s > 0$ and $|c| = (2\pi s^2)^{-1/4}$. Then $\langle Q\rangle = q$, $\langle P\rangle = p_0$, $\Delta Q = s$ and $\Delta P = \hbar/2s$.

Proof. First, the inequality. By Proposition 4.7, the vectors $Q\psi$ and $P\psi$ lie in $\mathcal{S}(\mathbb{R})$, so in $D(Q)$ and in $D(P)$. By the symmetry of Q and P and by Proposition 4.7,

$$\langle Q\psi, P\psi\rangle - \langle P\psi, Q\psi\rangle = \langle \psi, QP\psi - PQ\psi\rangle = \mathrm{i}\hbar.$$

Lemma 4.8 gives $\Delta Q\, \Delta P \geq \hbar/2$.

Next, equality. Write $\tilde{Q} = Q - q$ and $\tilde{P} = P - p_0$, with $q = \langle Q\rangle$ and $p_0 = \langle P\rangle$, and $z = \langle \tilde{Q}\psi, \tilde{P}\psi\rangle$. By the proof of Lemma 4.8, $z - \overline{z} = \mathrm{i}\hbar$, so $\operatorname{Im} z = \hbar/2$. By the same proof,

$$\Delta Q\, \Delta P \geq |z| \geq \operatorname{Im} z = \frac{\hbar}{2}.$$

Equality needs both steps to be equalities. The second gives $z = \mathrm{i}\hbar/2$. The first is equality in Cauchy–Schwarz. Since ψ is continuous and not 0, it is nonzero on an interval, and so is $(x-q)\psi(x)$. Hence $\tilde{Q}\psi \neq 0$, and equality in Cauchy–Schwarz gives $\tilde{P}\psi = \lambda\tilde{Q}\psi$ for a number λ. Then $z = \langle \tilde{Q}\psi, \lambda\tilde{Q}\psi\rangle = \lambda(\Delta Q)^2 = \mathrm{i}\hbar/2$, so $\lambda = \mathrm{i}\mu$ with $\mu = \hbar/2(\Delta Q)^2 > 0$. The equation $\tilde{P}\psi = \mathrm{i}\mu\tilde{Q}\psi$ reads

$$\psi'(x) = h'(x)\psi(x), \qquad h(x) = \frac{\mathrm{i}p_0 x}{\hbar} - \frac{\mu(x-q)^2}{2\hbar}.$$

So $(e^{-h}\psi)' = 0$, and $\psi = c\, e^{h}$ for a number c. With $\mu/2\hbar = 1/4s^2$, for a width $s > 0$, this is (4.3), and $\|\psi\| = 1$ fixes $|c|$.

Conversely, let ψ be (4.3). Then $|\psi(x)|^2 = (2\pi s^2)^{-1/2} e^{-(x-q)^2/2s^2}$, a normal density with the mean q and the spread s, so $\langle Q\rangle = q$ and $\Delta Q = s$. Differentiating, $-\mathrm{i}\hbar\psi' = p_0\psi + (\mathrm{i}\hbar/2s^2)(x-q)\psi$, so $(P - p_0)\psi = (\mathrm{i}\hbar/2s^2)(Q-q)\psi$. Since $\langle Q\rangle = q$, $\langle P\rangle = p_0 + (\mathrm{i}\hbar/2s^2)\,\langle \psi, (Q-q)\psi\rangle = p_0$ and $\Delta P = (\hbar/2s^2)\Delta Q = \hbar/2s$. The product is $\hbar/2$. ∎

Both spreads are finite exactly on $D(Q) \cap D(P)$, the natural domain of the inequality, and there it holds too, by a cited theorem.

Theorem 4.10 (Uncertainty on the natural domains · *cited*). *For every unit vector $\psi \in D(Q) \cap D(P)$, $\Delta Q\, \Delta P \geq \hbar/2$.*

Hall, Thm 12.7, proves it without assuming $\psi \in D(QP) \cap D(PQ)$. The equality case on these domains is cited too. A unit vector $\psi \in D(Q) \cap D(P)$ gives $\Delta Q\, \Delta P = \hbar/2$ exactly when ψ is one of the packets (4.3) (Hall, Props 12.10 and 12.11).

The line is essential to this theorem. Robertson's inequality holds in the box too. There the position, which we also write Q, multiplies by x on $L^2(0, L)$. Since $\|Q\psi\| \leq L\,\|\psi\|$, this Q

is bounded and defined on every vector. Since x is real, this Q is self-adjoint (*Linear Algebra*, Ex. 14.6). For the momenta P_θ of Definition 3.9 a boundary term enters.

Proposition 4.11 (The boundary term). *For every θ and every $\psi \in D(P_\theta)$,*

$$\langle Q\psi, P_\theta\psi\rangle - \langle P_\theta\psi, Q\psi\rangle = \mathrm{i}\hbar\big(\|\psi\|^2 - L\,|\psi(L)|^2\big).$$

Proof. By the product rule for $\mathcal{A}$ (§ 3.2), $Q\psi$ lies in $\mathcal{A}$, with $(Q\psi)' = \psi + x\psi'$, but in general not in $D(P_\theta)$. By (3.2) with $\varphi = Q\psi$,

$$\langle Q\psi, -\mathrm{i}\hbar\psi'\rangle - \langle -\mathrm{i}\hbar(\psi + x\psi'), \psi\rangle = -\mathrm{i}\hbar\, L\,|\psi(L)|^2\,,$$

since $x = 0$ at the left wall. The first inner product is $\langle Q\psi, P_\theta\psi\rangle$. Since x is real, the second is $\mathrm{i}\hbar\,\|\psi\|^2 + \langle P_\theta\psi, Q\psi\rangle$. Moving $\mathrm{i}\hbar\,\|\psi\|^2$ to the right side gives the claim. ∎

So for a unit vector $\psi \in D(P_\theta)$, Robertson's inequality gives $\Delta Q\,\Delta P_\theta \geq (\hbar/2)\left|1 - L\,|\psi(L)|^2\right|$. For the eigenvector $e_{\theta,k}$ of Theorem 3.10, $L\left|e_{\theta,k}(L)\right|^2 = 1$, and indeed $\Delta P_\theta = 0$ while $\Delta Q = L/\sqrt{12}$ is finite. So the product is 0, and $\Delta Q\,\Delta P_\theta \geq \hbar/2$ fails at the eigenvectors of every P_θ. On $D(P_D)$, where $\psi(L) = 0$, the inequality holds. The line has no walls and no boundary term.

The spreads belong to separate runs, as in *Quantum Mechanics I*, Lecture 5. On some copies of one preparation the position is read, and on others the momentum. Theorem 4.9 says that no preparation makes both spreads small. It says nothing about a position reading disturbing a later momentum reading.

Self-check 4.4

1. In the box, compute $\Delta Q\,\Delta P_D$ for u_1 and compare it with $\hbar/2$.

4.5 A Gaussian's momenta

The momentum readings of a Gaussian packet are found from the transform of (4.3).

Example 4.12 (The momentum distribution of a Gaussian packet). Take the packet (4.3) with $c = (2\pi s^2)^{-1/4}$.

1. *A Gaussian integral.* For real k, put

$$G(k) = \int e^{-\mathrm{i}ky}\, e^{-y^2/4s^2}\,\mathrm{d}y.$$

We differentiate under the integral by a Taylor bound. Since $\left|e^{-\mathrm{i}hy} - 1 + \mathrm{i}hy\right| \leq h^2y^2/2$ for real h, the difference quotient $(G(k+h)-G(k))/h$ differs from $-\mathrm{i}\int e^{-\mathrm{i}ky}\, y\, e^{-y^2/4s^2}\,\mathrm{d}y$ by at most $(|h|/2)\int y^2 e^{-y^2/4s^2}\,\mathrm{d}y$, which tends to 0 as $h \to 0$. Since $ye^{-y^2/4s^2} = -2s^2(e^{-y^2/4s^2})'$, integration by parts gives

$$G'(k) = -\mathrm{i}\int e^{-\mathrm{i}ky}\, y\, e^{-y^2/4s^2}\,\mathrm{d}y = -2s^2k\,G(k).$$

So $(e^{s^2k^2}G)' = 0$, and $G(k) = G(0)e^{-s^2k^2}$. The equation does not fix the constant. We have $G(0) = \int e^{-y^2/4s^2}\,\mathrm{d}y = 2s\sqrt{\pi}$, by the substitution $y = 2st$ and $\int e^{-t^2}\,\mathrm{d}t = \sqrt{\pi}$. Hence

$$G(k) = 2s\sqrt{\pi}\, e^{-s^2k^2}.$$

2. *The transform.* Substitute $x = q + y$ in (4.1). The phase separates as

$$e^{-\mathrm{i}(p-p_0)q/\hbar} e^{-\mathrm{i}(p-p_0)y/\hbar}.$$

   So the integral of the first step appears, with $k = (p - p_0)/\hbar$:

$$\tilde{\psi}(p) = \frac{c}{\sqrt{2\pi\hbar}}\, e^{-\mathrm{i}(p-p_0)q/\hbar}\, G\Big(\frac{p-p_0}{\hbar}\Big).$$

   The constants combine to

$$\tilde{\psi}(p) = \Big(\frac{2s^2}{\pi\hbar^2}\Big)^{1/4} \exp\Big(-\frac{s^2(p-p_0)^2}{\hbar^2} - \frac{\mathrm{i}(p-p_0)q}{\hbar}\Big).$$

3. *The distribution.* With $\sigma = \hbar/2s$,

$$\big|\tilde{\psi}(p)\big|^2 = \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-(p-p_0)^2/2\sigma^2},$$

   a normal density with the mean p_0 and the spread $\hbar/2s$, as Theorem 4.9 found. A momentum reading falls within $\hbar/2s$ of p_0 with probability 0.683, and within $\hbar/s$ with probability 0.954. The narrower the packet in position, the wider its momentum distribution.
4. *An electron.* An electron in such a packet, with the width $s = 0.1$ nm of an atom, has the velocity spread $\Delta P/m_\mathrm{e} = \hbar/2m_\mathrm{e}s \approx 5.8 \times 10^5$ m/s.

A Gaussian density of positions does not make $\Delta Q\, \Delta P$ equal to $\hbar/2$, since the phase matters. For a real $b \neq 0$, the *chirped Gaussian*

$$\psi_b(x) = (2\pi s^2)^{-1/4} \exp\Big(-\frac{(1+\mathrm{i}b)x^2}{4s^2}\Big)$$

has $|\psi_b| = |\psi_0|$, so $\Delta Q = s$. Differentiating, $-\mathrm{i}\hbar\psi_b' = \mathrm{i}\hbar(1+\mathrm{i}b)x\psi_b/2s^2$. Since $\int x\,|\psi_b|^2\,\mathrm{d}x = 0$, the mean of P is 0, and

$$(\Delta P)^2 = \|P\psi_b\|^2 = \frac{\hbar^2(1+b^2)}{4s^4}\int x^2\,|\psi_b|^2\,\mathrm{d}x = \frac{\hbar^2(1+b^2)}{4s^2}.$$

So $\Delta Q\, \Delta P = (\hbar/2)\sqrt{1+b^2}$, which exceeds $\hbar/2$. A free packet acquires such a phase as it spreads (Lecture 7).

Figure 4.2 draws the position and momentum densities at $\hbar = 1$ for the packets (4.3) with $q = p_0 = 0$ and the widths $s = 1$ and $s = 1/2$, and for the chirped Gaussian with $s = 1/2$ and $b = 2$. Halving s halves the spread of the positions and doubles the spread of the momenta. The chirp leaves the position density unchanged, so its dashed curve lies on that of $s = 1/2$. It widens the momentum density by the factor $\sqrt{1+b^2} = \sqrt{5}$.

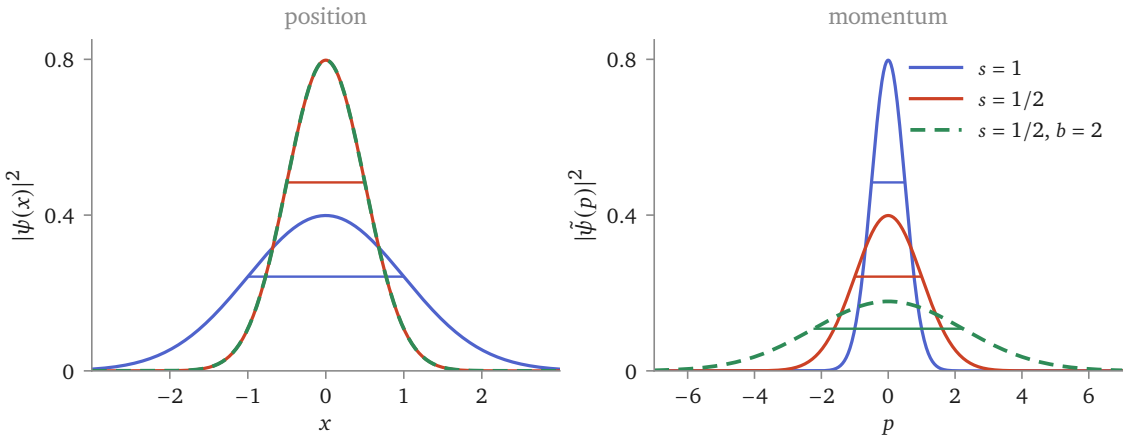


Figure 4.2: The position densities $|\psi(x)|^2$ and the momentum densities $|\tilde{\psi}(p)|^2$ of three packets with $q = p_0 = 0$, at $\hbar = 1$, each in one colour in both panels. A horizontal bar spans one spread on each side of the mean.

Self-check 4.5

1. Take (4.3) with $p_0 = \hbar/s$. What is the probability that a momentum reading is negative?

The physics

De Broglie proposed in 1924 that a particle of momentum p has the wavelength h/p, with $h = 2\pi\hbar$. A packet whose momentum density is concentrated near p_0 oscillates as $e^{ip_0x/\hbar}$, with this period. In 1999 Arndt, Nairz, Vos-Andreae, Keller, van der Zouw and Zeilinger sent molecules of C_{60}, of mass 720 u, through a grating of period 100 nm at about 220 m/s. Their wavelength is about 2.5 pm, some 400 times smaller than the molecule's diameter. So the first diffraction maxima lie about 25 μrad from the beam, and the experiment resolved them.

The free particle

Observables: the position Q, the momentum $P = F^{-1}M_pF$, with F the Fourier transform (Theorem 4.2), and the free energy $H_0 = P^2/2m$ are self-adjoint, and none has an eigenvalue. The densities $|\psi|^2$ and $|\tilde{\psi}|^2$ give the readings of Q and P on intervals (Theorem 4.6 and Postulate 4.5). A plane wave is not a state. *The canonical relation:* $QP\psi - PQ\psi = i\hbar\psi$ for every $\psi \in D(QP) \cap D(PQ)$ (Proposition 4.7). *Uncertainty:* for unit $\psi \in \mathcal{S}(\mathbb{R})$, $\Delta Q\, \Delta P \geq \hbar/2$, with equality exactly for the Gaussian packets (4.3), whose momentum density is normal with the spread $\hbar/2s$ (Theorem 4.9 and Example 4.12). In a box the inequality fails for P_θ, at its eigenvectors (Proposition 4.11).

Summary.

- *The transform of a derivative*: on $\mathcal{S}(\mathbb{R})$, the transform of $-i\hbar\psi'$ is $p\tilde{\psi}$ (Proposition 4.3), and Plancherel makes the transform unitary (Theorem 4.2).
- *Position, momentum and free energy* are self-adjoint, none has an eigenvalue, and $H_0 = P^2/2m$ (Theorem 4.6).
- *The canonical commutation relation* holds on $\mathcal{S}(\mathbb{R})$ and on $D(QP) \cap D(PQ)$ (Proposition 4.7).
- *The uncertainty relation*: $\Delta Q\, \Delta P \geq \hbar/2$ for unit $\psi \in \mathcal{S}(\mathbb{R})$, with equality exactly for the Gaussian packets (Theorem 4.9), and on $D(Q) \cap D(P)$ by a cited theorem (Theorem 4.10). In a box, the boundary term breaks it for P_θ, at its eigenvectors, and it holds on $D(P_D)$ (Proposition 4.11).
- *The momentum distribution of a Gaussian packet* is normal, with the spread $\hbar/2s$ (Example 4.12).

Lecture 5 puts the readings of Q and P on intervals into one rule for every self-adjoint operator, the spectral measure.

Exercises. Exercise Sheet 4.

Self-check answers.

4.1 1. Substitute $y = x - a$, and use $e^{-ipx/\hbar} = e^{-ipa/\hbar}e^{-ipy/\hbar}$. For the second, $e^{-ipx/\hbar}e^{ibx/\hbar} = e^{-i(p-b)x/\hbar}$.

4.2 1. Yes, since $\int (1+x^2)^{-3/2}\,dx = 2$. No, since $x^2(1+x^2)^{-3/2}$ behaves like $1/|x|$ for large $|x|$, so $\int |x\psi|^2$ diverges.

4.3 1. $\psi' = -(x/2s^2)\psi$ and $\int x^2\,|\psi|^2 = s^2$, so $\|P\psi\|^2 = \hbar^2/4s^2$ and $\langle\psi, H_0\psi\rangle = \hbar^2/8ms^2$. It tends to 0 as $s \to \infty$. No state has the mean energy 0, since $\int (p^2/2m)\,|\tilde{\psi}|^2\,dp = 0$ forces $\tilde{\psi} = 0$. So a free particle has no ground state.

4.4 1. $\langle Q\rangle = L/2$, $(\Delta Q)^2 = L^2(1/12 - 1/2\pi^2)$ and $\Delta P_D = \pi\hbar/L$, so $\Delta Q\, \Delta P_D = \pi\hbar\sqrt{1/12 - 1/2\pi^2} \approx 0.568\,\hbar$, above $\hbar/2$.

4.5 1. The momentum density is normal with the mean $\hbar/s$ and the spread $\hbar/2s$, so a negative reading lies more than two spreads below the mean: probability 0.023.

References

Teta, *A Mathematical Primer on Quantum Mechanics*, §6.1 (pp. 139–140). Hall, *Quantum Theory for Mathematicians*, §12.2 (p. 245), Thm 12.7 (p. 246), Props 12.10 and 12.11 (pp. 249–250), Thms A.18 and A.19 (pp. 532–533), Prop. A.22 (p. 533). Galindo and Pascual, *Quantum Mechanics I*, §3.4, (3.39)–(3.42) (pp. 95–96). *Quantum Mechanics I*, Thm 5.7 and Prop. 9.8. *Linear Algebra for Physics*, Lecture 13, Thm 14.15, Ex. 14.6. Arndt, Nairz, Vos-Andreae, Keller, van der Zouw and Zeilinger, *Nature* 401, 680 (1999).

Galindo and Pascual, (3.39)–(3.42), find the same packets as Theorem 4.9. Hall, Prop. 12.10, writes the equality case on $D(Q) \cap D(P)$ as the equation $(Q + i\delta P)\psi = \lambda\psi$, for a nonzero real δ and a complex λ. His Prop. 12.11 solves this equation, with the derivative taken in a weaker sense. Hall, §12.2, gives the same example of the box as Proposition 4.11. Hall, Prop. A.22, states the integral G of Example 4.12.